



Annual Maintenance Completion Report

2025-2026

Generating unit III



Mechanical Maintenance Unit

Tala Hydropower Plant

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1. Overview

The alignment of the unit during this annual maintenance was done purely based on the recommendations and report of the trend on conditioning & monitoring of data such as system temperature and abnormal vibration provided by the HR&DC.

The major work plan in this unit included nozzle injector overhauling based on condition-based oil pressure testing, Replacement of Pelton runner 7020, overhauling of governor OPU accumulator for replacement of cup seal, replacement of Emergency water pipeline, sump cleaning etc.

Scheduled vs Actual date of start and completion of maintenance on the generating unit is shown in Table-1

UNIT-III Annual Maintenance

The Annual Maintenance work of the unit commenced on March 04. 2026 but due to the Replacement of Service Seal in unit 3 the work is completed only 22 April and dry run was initiated the machine was finally and successfully synchronized at 16 hours on 22nd April.

The major maintenance activities conducted for the unit are detailed hereunder and all test/calibration reports, checklists and inspection notes are placed as Annexure.

TABLE 1: AM SCHEDULE FOR GENERATING UNIT III

SN	Unit #	Approved Schedule			Actual Completion		
		Start	End	No of Days	Start	End	No of Days
1.	Unit III	04.03.2026	30.04.2026	50	04.03.2026	22.04.2026	50

All major work has completed on 20.04.2026. but unit was .

2. Turbine

TABLE 2 LIST OF SPARES CONSUMED FOR TURBINE SYSTEM

S. No	Materials	Quantity
1	Cup seal p.p200x230x16mm	2 no
2	Wiper Seal PU with casing 200x220x12/16mm,	2 no
3	PTFE ring seal 200x209x2x7.5mm with inner	2 NOS
4	Cup seal p. p 110x130x12mm	4no
5	Wiper Seal PU with casing 110x120x7/10mm	4 no
6	NBR endless O-ring 335x6mm 70 shore	4
7	NBR Endless o-ring 320x6mm 70 shore	16
8	NBR endless o-ring 246x6mm 70 shore	6
9	NBR endless o-ring 19.5x3mm 70 shore	8
10	H. T Allen bolt, M20x150 H/Thread, Gr 12.9	5
11	H. T Allen bolt 16x160 grade 12.9	2
12	NBR Endless o-ring 28x3mm 70 shore	2
13	Ss stud 12x307 with nut and washer of MCD13	22

14	SS Hex headed bolt and Nut M10x60mm half	32
15	SS Allen bolt M36x340 grade 9.8	44
16	SS Allen bolt M20x200 grade 9.8	3

2.1. Nozzle Injector Assembly

All the five nozzle injector assemblies were subjected to functional check with in-situ oil pressure testing. During the testing with an initial pressure of 100 bar for time duration of 30 minutes, nozzle J1 And J5 was found to have heavy water leakage and maximum pressure drop was observed in nozzle J1. Therefore, both nozzles J1, and j5 were dismantled for overhauling during the 2025-2026 annual maintenance. Those nozzle injector assemblies that did not require to be dismantled had their needle tip and liner assemblies overhauled and thoroughly cleaned in-situ. The needle tip-holding bolts were replaced. While overhauling, all the internal fasteners were replaced with new set.



FIGURE 1 UNDER WATER COMPONENT INSPECTION WHILE OVERHAULING THE NOZZLE INJECTOR SUB ASSEMBLIES

Due diligence was given while tightening the fastener ensuring equal tightening torque is provided. All the critical assemblies of these components were carried out under the supervision of the concerned engineers. The checklist that are maintained and recorded are attached as annexure.

2.2 Runner

TABLE 3 RUNNER PARTICULARS

Particulars	Runner (7020)
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Make	Andritz Hydro, Switzerland
Type	Forged Fabricated and hard coated
Hard Coating	HVOF
Cumulative Running Hours	42797.42hours as of 24.04.2026
Cumulative running after repair	6730.37 hours as of 24/04/2026

The unit III runner 7020 was in operation from 6730.37 and its running hour since the commission has reached 6730.37 hours by the time the unit was taken under shutdown for annual maintenance 2025-2026. The physical condition of the runner bucket profile is observed Erosion. The Teams has carried out repair of Bucket. The NDT personnel from HR&DC examined the physical condition of all the runner buckets after visual inspection followed by Non-destructive test viz. Magnetic particle testing and Dye penetration test (DPT) on the bucket roots and eroded parts on all buckets with the no wear. Though few pitting were revealed from the test, no severe indication was detected on buckets. MMU has decided to not replace the runner. The runner will be inspected after 1500 hours as per schedule, and the decision for its replacement will be taken up accordingly.

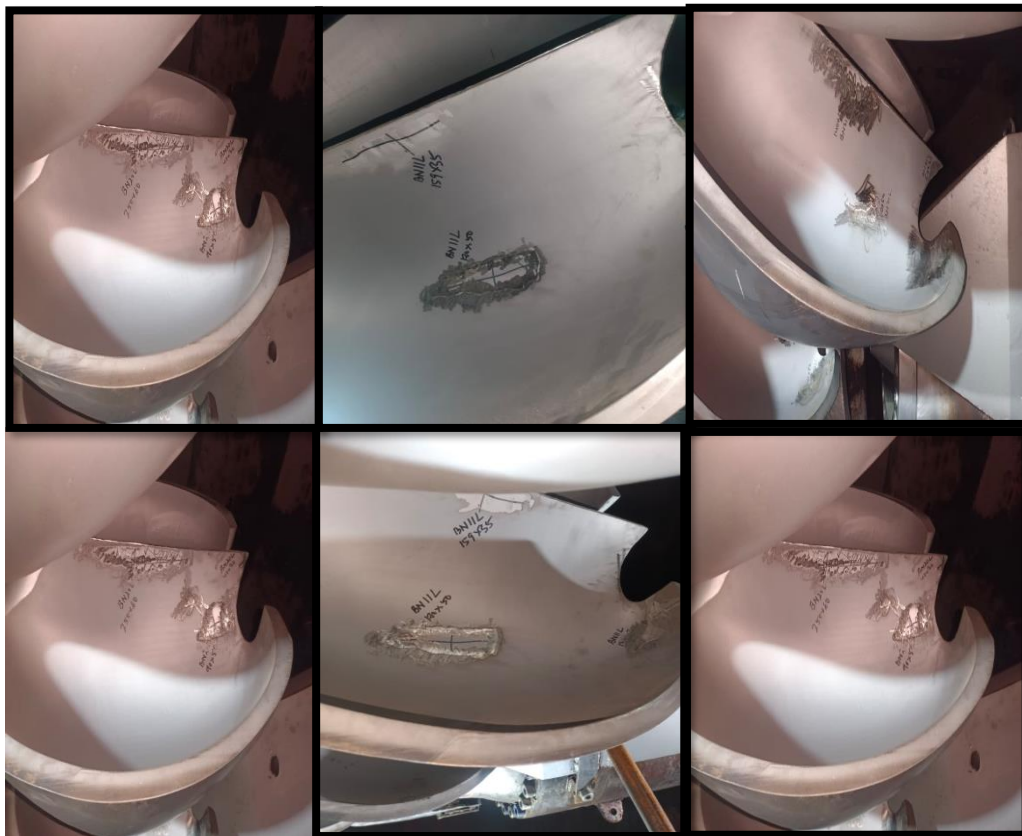
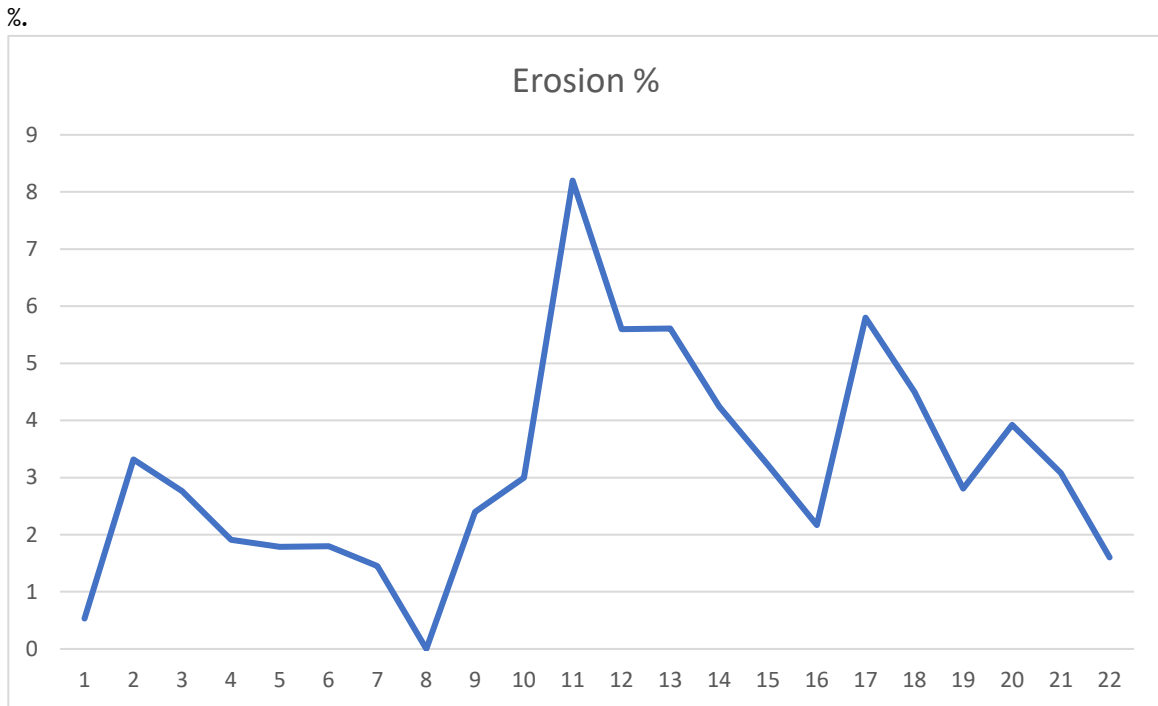


FIGURE 2 NDT (VISUAL, DPT AND MPT) INSPECTION OF MICRO GAUSS RUNNER 7020

The below chart gives information on parent material erosion on each bucket of the runner. The erosion is not consistent with maximum 8.2 % estimated to be eroded that was witnessed on bucket 11 while the least erosion is observed on buckets 1, and 8 whose erosion Less than 1%.



3. Governor and MIV System

TABLE 4 SPARES CONSUMED FOR GOVERNOR AND MIV SYSTEM

S. No	Description	Quantity
1	Filter element-0160 DNo10BN/HC	2 no
2	Governor OPU return Line duplex filter	2 no
3	Filter element0160 D 020 BN4HC	1 no

3.1 Governing System

The main function of governor system is to maintain the rotational speed of turbine within acceptable limits during no-load condition. It performs this function by sensing changes in speed and accordingly controlling flow of water into turbine at fast rate to maintain the speed with reference to grid frequency.

For this unit, the SCADA implementation had been successfully completed during the previous Annual Maintenance. Accordingly, no works related to the HMC or NCC were required this year, unlike in past maintenance cycles where such activities were a major part

of the scope. Instead, routine servicing tasks were carried out to ensure system reliability. These included thorough cleaning of the OPU tank, inspection of pump and oil filters, and replenishment of nitrogen pressure in the accumulator bank to maintain optimal system performance. All high-pressure pipelines associated with VTA (Valve Table Assembly) were carefully inspected for any signs of leakage, and necessary tightening or corrective measures were taken wherever required.

Valve Table Assembly (VTA)

The VTA is a critical component of the hydraulic control system, primarily responsible for directing and regulating oil flow to the servomotors of the needle and deflector systems. It houses the solenoid valves, pressure gauges, and other control elements that interface with the SCADA system for remote operation and monitoring.

With the SCADA integration completed during the previous Annual Maintenance, the VTA now facilitates more precise and efficient control of the turbine's hydro-mechanical functions, reducing the need for manual intervention and improving overall system responsiveness.



Figure 3 VTA system

TABLE 5 DEFLECTOR GAP SETTING READING

Nozzle	Initial reading (mm)	Final reading (mm)
J1	7.4	7.4
J2	16	8
J3	12	8
J4	9	8
J5	9	8

Jockey pump



FIGURE 4 jockey pump

In a hydro power plant a jockey pump is used to maintain the oil pressure in the hydraulic system of the turbine governor system. This is important for ensuring the proper functioning and control of the turbine during start-up, shutdown and load changes. The jockey oil pump helps to prevent damage to the turbine components and ensure reliable operation of the governing system.

3.2 MIV System

The MIV system is the primary isolation for turbine and principal protection for the station against penstock water. It has a spherical valve with two-part body, door, a downstream service seal and upstream maintenance seal. The valve is operated by a double acting oil operated servomotor which is controlled by a hydraulic operated valve.

The N₂ gas for MIV N₂ bank has been topped up and tested for N₂ gas leakage with surf foam bubble, arresting the leakage by replacing O-ring in the main outlet valve. The N₂ gas piston accumulator for Governor OPU was checked and replenished to maintain the minimum pressure of 60 kg/cm².

The pressure gauges of MIV control panel and accumulator have been calibrated at HRDC and three of them viz. maintenance seal opening, penstock pressure and service seal opening gauges have failed during the calibration and testing.



FIGURE 5 NITROGEN REPLENISHMENT IN ACCUMULATORS/ REPLACEMENT OF PRESSURE GAUGES AFTER CALIBRATION

3.2.1. MIV OPU

OPU pump capacity was checked and found normal. Functionality check for internal leakages and servicing was carried on all the directional control solenoid valves provided in OPU and MIV control panel. The MIV lubricating oil was drained and the oil sump was cleaned thoroughly. The reconditioned oil was replenished again. All setting and dry operation on MIV system was checked and verified from the control panel. Contactors and limit switches of MIV were checked for their healthiness and replacement done wherever necessary. During the functionality check, no N₂ gas leakage was observed.



FIGURE 6 DRAINING OF MIV OPU TANK OIL AND CLEANING

3.2.2. Decompression valve, bypass valve and pressure accumulators

In order to ensure safe parameters of the control pipelines for safe and smooth operations, it is highly recommended to perform non-destructive testing periodically since these are considered among the most sensitive parts.

The ultrasonic testing of MIV bypass pipeline, pressure accumulator and distributor pipeline for thickness verification was done by the NDT personnel from HRDC to inspect any possible deviations in the pipeline thickness or any crack development before it amplifies.



FIGURE 7 miv system

TABLE 6 SPARES CONSUMED FOR Auxiliary system

S. No	Description	Quantity
1	Hex bolt-Pm16x50-8.8R16drg12020160701	18 no
2	Drive end mechanical seal with gland	2 no
3	NDE Mechanical seal with gland	1 no
4	Bearing SKF 6309 2z/C3	2 no
5	Wearing ring CF8m, part no 190.0	1 no
6	Bearing housing D. E, part no240.1	1 no
7	SS ball valve 50NBwith electric actuator	1 no

4.1. Clean Cooling Water System

Two circulating pumps, one main and other standby, are provided for pumping clean water for cooling of the generator stator, bearing coolers, and the lubrication system. During Annual Maintenance, the pump components were dismantled for servicing and inspection. The mechanical seals, bearings and glands and shaft sleeve were replaced with new sets in Pump A. The functionality test and thorough servicing of motors and pumps were carried out. The motor casing was scrapped and painted in order to avoid corrosion. Besides, the non-return valve (NRV) was also serviced. The alignment work for both the pumps A and B was carried out after completion of servicing and assembly. After rectifications, the pumps were checked for operation and found healthy.



FIGURE 8 OVERHAULING AND SERVICING OF CCW PUMPS AND REPLACEMENT OF RADIAL BALL BEARINGS

Cleaning local control panels, contactors and necessary replacement were done wherever required. Tightness and overall healthiness were checked with control test of the complete system and it was found normal and healthy. The terminations were checked for looseness and tightened accordingly.

4.2. Raw Cooling Water System

Two vertical turbine pumps are provided in each unit for cooling of clean water in Heat Exchangers, and to cool down the generator transformer and shunt reactor of the unit. During maintenance period, the NRVs of both the RCW pumps of the unit were inspected and serviced accordingly. Functionality of motors and pumps was checked and serviced. After servicing, motor-pump shaft alignment was carried out.

Figure 8



FIGURE 9 CLEANING AND FLUSHING OF CYCLONE SAND SEPARATOR

Motorized backwash strainer which is used to trap the coarse sediments from the raw water was overhauled, thoroughly cleaned and assembled back. The filter elements were removed and replaced with a spare set. Visual inspection, thorough cleaning and replacement of lubrication oil were done. The foundation bolts were checked for looseness and grease applied wherever necessary, Installation of Actuator in cyclone separator. After servicing, the pump-motor and the strainer were functioning normally. The cyclone sand separator was also cleaned and rinsed. The sediments collected at the bottom during operations were taken out manually and then flushed

Cleaning local control panels, contactors and necessary replacement were done wherever required. Tightness and overall healthiness were checked with control test of the complete system and it was found normal and healthy. The terminations were checked for looseness and tightened accordingly. Install new pressure gauge and digital gauge.

4.3. Heat Exchanger

A heat exchanger (MHE) is provided in each unit in order to circulate clean cooling water in the generating system for the cooling of stator air coolers, OPU coolers and bearing coolers, with the help of raw water supplied into the tubes inside the heat exchanger. The shell-and-tube type heat exchanger was opened, the tubes were thoroughly cleansed and then flushed out, Replace the out let Flowmeter in MHE. The pressure testing of the heat exchanger was carried out after cleaning and fitting back of end covers. The test result is attached as annexure.



FIGURE 10 SERVICING AND PRESSURE TESTING OF MAIN HEAT EXCHANGER

5. Internal/ external services availed

- Runner inspection (Magnetic Particle Inspection) by HRDC
- Ultrasonic testing on decompression, bypass pipelines, pressure accumulators and distributor pipeline joints by HRDC

6. Maintenance Challenges to be recorded

- During the pre-testing of nozzles, nozzles J1, and J5 were found to have heavy water leakage. Therefore, the Two nozzles had to be dismantled and overhauled for thorough servicing and inspection
- Replacement of Runner in unit 5
- Excitation issue (resolved by CPU)

7. Any Modification and improvement

- Carbon steel emergency cooling water pipelines replacement MS pipe.
- Connected the jockey pump.
- Replacement of MIV seal in unit 2, 3 and 6.
- Dragging of sand at RD60 and RD200
- Replacement of Runner in Unit V

Submitted



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Mechanical maintenance unit

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